Ayush Kumar Tarun

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Education

Ph.D., Computer and Communication Sciences Sep 2024 – Present
EPFL, Lausanne, Switzerland
– Advisor: Prof. Antoine Bosselut (NLP Lab). EPFL AI Center PhD Fellow (2026–27).

B.E. Computer Science, Minor in Data Science 2018 – 2022
BITS Pilani, Pilani, India
– CGPA 9.30/10. Merit scholarships for ranking in the top 1–3% of the batch throughout.

Selected Publications

* denotes equal contribution. Full list on Google Scholar.

1. Overcoming Data Imbalances with Expert Selection Biases
   A. K. Tarun, C. Meister, A. Bosselut. **Preprint, 2026**.
2. Scaling Laws for Multimodal Data Mixtures
   A. Khandelwal*, **A. K. Tarun***, Y. Xu, I. Schlag, G. Farnadi, S. Reddy, L. Zettlemoyer, A. Bosselut. **NeurIPS 2026 (Spotlight)**.
3. Helpful to a Fault: Measuring Illicit Assistance in Multi-Turn, Multilingual LLM Agents
   N. Talokar*, **A. K. Tarun***, M. Mandal, M. Andriushchenko, A. Bosselut. **ICML 2026**.
4. Apertus: Democratizing Open and Compliant LLMs for Global Language Environments
   Project Apertus. **ACL 2026**.
5. Chitrarth: Bridging Vision and Language for a Billion People
   S. Khan, **A. K. Tarun**, A. Ravi, A. Faraz, P. K. Pokala, A. Bhangare, R. Kolla, C. Khatri, S. Agarwal. **ICASSP 2025**.
6. EcoVal: An Efficient Data Valuation Framework for Machine Learning
   A. K. Tarun*, V. S. Chundawat*, M. Mandal, H. M. Tan, B. Chen, M. Kankanhalli. **KDD 2024**.
7. Deep Regression Unlearning
   A. K. Tarun*, V. S. Chundawat*, M. Mandal, M. Kankanhalli. **ICML 2023**.
8. Can Bad Teaching Induce Forgetting? Unlearning in Deep Networks Using an Incompetent Teacher
   V. S. Chundawat*, **A. K. Tarun***, M. Mandal, M. Kankanhalli. **AAAI 2023**.
9. Fast Yet Effective Machine Unlearning
   A. K. Tarun*, V. S. Chundawat*, M. Mandal, M. Kankanhalli. **IEEE TNNLS 2023**.
10. Zero-Shot Machine Unlearning
    V. S. Chundawat*, **A. K. Tarun***, M. Mandal, M. Kankanhalli. **IEEE TIFS 2023**.

Research Experience

Doctoral Researcher Oct 2024 – Present
EPFL NLP Lab / Swiss AI Initiative, Lausanne, Switzerland
– **Apertus-2:** Throughput optimisation of the parallelism configuration and all-to-all communication for the pretraining of a 500B+ parameter Mixture-of-Experts model on the Alps supercomputer (CSCS), and mitigation of data imbalance through interventions in expert routing.
– Derived scaling laws for multimodal data mixtures; designed and supervised the synthetic-data pipeline for multilingual audio; developed multi-turn, multilingual safety evaluations for LLM agents.
– **Apertus:** Key contributor to the post-training (alignment) of Switzerland's open, compliant 70B LLM.

Research Assistant Oct 2022 – Feb 2024
N-CRiPT, National University of Singapore, Singapore
– Proposed EcoVal (KDD 2024), an efficient data-valuation framework that is ∼1000× faster than Shapley-style baselines on 2k samples.
– Developed a family of machine-unlearning methods for deep networks: regression unlearning (ICML 2023), distillation from an incompetent teacher (AAAI 2023), and error-maximising unlearning that is 150× faster than prior work (TNNLS). Introduced and formalised the problem of *zero-shot machine unlearning* (TIFS).

Industry Experience

Research Engineer II Feb 2024 – Sep 2024
Krutrim, Bangalore, India
- Member of the team that built Krutrim-1 and Krutrim-2, India's first open-source sovereign LLMs.
- Built vision-language models for 22 Indian languages: the generalist Chitrarth and the multimodal-translation model Chitranuvad.
- Shipped web-grounded retrieval-augmented generation (RAG) for the Krutrim chat product. Also contributed to training Indic embedding models for multilingual retrieval and semantic search.

Principal Applied Scientist Jul 2023 – Jan 2024
Famdo, Bangalore, India
- Single-handedly built v1 of an LLM-powered assistant for a doctor–patient platform that automatically answers patients' text and voice queries in their local language.
- Fine-tuned Famdo's proprietary healthcare LLM with QLoRA/PEFT and deployed it end to end on AWS ECS and SageMaker.

ML Engineer → Applied Scientist Jan 2022 – Jun 2023
Pepper Content, Mumbai, India
- Built in-house Indic neural machine translation (NMT) models that launched the company's automated localisation product line ($1M first deal).
- Built an embedding-based creator-recommendation system that cut mean allocation time from 29 h to 1.5 h, a deadline-risk predictor that raised on-time delivery by 52%, and a client-sentiment model that reduced churn by 35%.

Honours & Fellowships

EPFL AI Center PhD Fellowship — competitive fellowship awarded to outstanding EPFL doctoral researchers pursuing high-impact AI research. 2026 – 2027
BITS Pilani Merit Scholarship — 100% tuition waiver (top 1%), then 80% (top 3%). 2018 – 2022

Skills

Languages	Python, SQL; familiar with C, C++, CUDA, JavaScript, Java, R
ML / Systems	PyTorch, Megatron-LM, distributed training (FSDP/TP/PP/EP), HuggingFace, vLLM
Infrastructure	Slurm, AWS (ECS, SageMaker), Linux, Git